

PROJECT DEVELOPMENT PHASE

Electric Vehicle Analytics Dashboard using Flask & Tableau E-CarStart

Project Information

Project	Electric Vehicle Analytics Dashboard
Website	E-CarStart
Frontend	HTML, CSS, Bootstrap
Backend	Python Flask
Visualization	Tableau Public
Version Control	Git & GitHub

Introduction

This phase focuses on implementing the Electric Vehicle Analytics Dashboard. The web application was developed using Flask for backend routing, Bootstrap for a responsive interface, and Tableau Public for interactive data visualization.

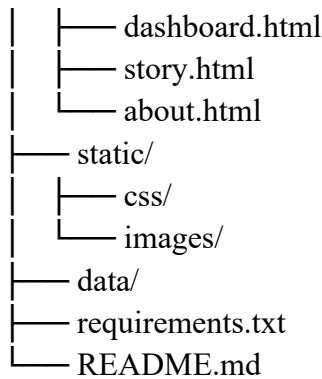
Development Environment

Programming Language: Python
IDE: Visual Studio Code
Framework: Flask
Visualization Tool: Tableau Public
Version Control: Git & GitHub
Browser: Google Chrome

Project Folder Structure

Electric-Vehicle-Analytics/

```
|
|— app.py
|— templates/
|   |— index.html
```



Module Development

- Home Module: Displays project overview and navigation.
- Dashboard Module: Embeds the Tableau Public dashboard.
- Story Module: Displays the Tableau story.
- About Module: Provides project information.
- Navigation Module: Enables seamless movement across pages.

Flask Implementation

Flask routes were created for the Home, Dashboard, Story, and About pages. HTML templates are rendered dynamically using Flask's `render_template()` function.

Frontend Development

The frontend was designed using HTML5, CSS3, and Bootstrap 5 to provide a clean, responsive, and user-friendly interface across different screen sizes.

Tableau Integration

The Tableau Public dashboard and story were embedded into the Flask application using Tableau's JavaScript API, enabling users to interact with visualizations directly from the website.

Testing During Development

Each module was tested individually to verify navigation, responsiveness, dashboard loading, story loading, image rendering, and overall functionality.

Challenges Faced

- Resolving Tableau embedding issues.
- Fixing image path problems.
- Improving UI responsiveness.
- Ensuring consistent navigation between pages.

Project Outcome

The completed application successfully integrates Flask with Tableau Public to provide an interactive Electric Vehicle Analytics platform with a responsive web interface.

Conclusion

The development phase transformed the project design into a functional web application. By integrating Flask and Tableau, the project delivers an intuitive platform for exploring EV analytics.